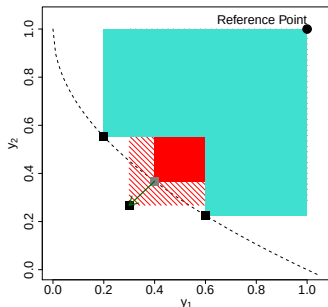
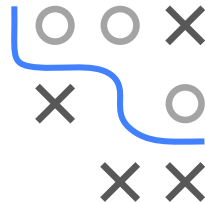


Optimization in Machine Learning

Bayesian Optimization

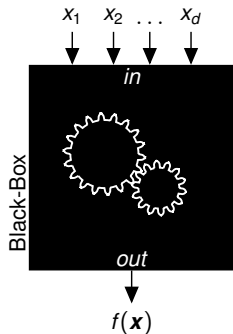
Multicriteria Bayesian Optimization



Learning goals

- Multicriteria Optimization
- Taxonomy
- ParEGO, SMS-EGO, EHI

MULTICRITERIA BAYESIAN OPTIMIZATION



$$f : \mathcal{S} \rightarrow \mathbb{R}^m$$

$$\min_{\mathbf{x} \in \mathcal{S}} f(\mathbf{x}) = (f_1(\mathbf{x}), \dots, f_m(\mathbf{x}))$$

- A configuration $\mathbf{x}$ **dominates** ($\prec$) $\tilde{\mathbf{x}}$ if

$$\forall i \in \{1, \dots, m\} : f_i(\mathbf{x}) \leq f_i(\tilde{\mathbf{x}})$$

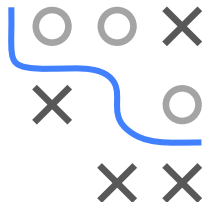
and $\exists j \in \{1, \dots, m\} : f_j(\mathbf{x}) < f_j(\tilde{\mathbf{x}})$

- Set of non-dominated solutions:

$$\mathcal{P} := \{\mathbf{x} \in \mathcal{S} \mid \nexists \tilde{\mathbf{x}} \in \mathcal{S} : \tilde{\mathbf{x}} \prec \mathbf{x}\}$$

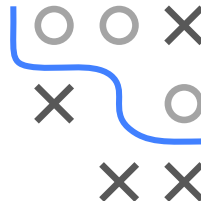
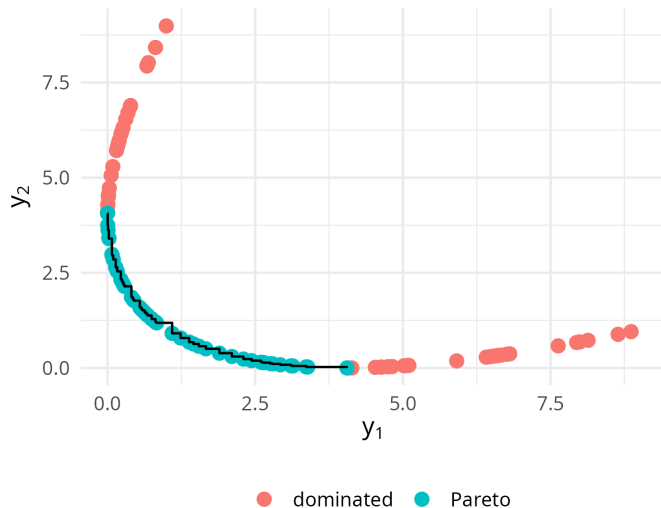
- Pareto set $\mathcal{P}$, Pareto front $\mathcal{F} = f(\mathcal{P})$

- Goal: Find $\hat{\mathcal{P}}$ of non-dominated points that estimates the true Pareto set $\mathcal{P}$



MULTICRITERIA BAYESIAN OPTIMIZATION

Example Pareto front:



MULTICRITERIA BAYESIAN OPTIMIZATION

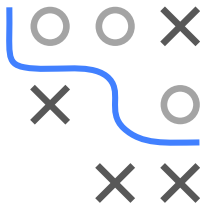
The most popular quality indicator is the hypervolume indicator (also called dominated hypervolume or $\mathcal{S}$ -metric).

The hypervolume, HV, of an approximation of the Pareto front $\hat{\mathcal{F}} = f(\hat{\mathcal{P}})$ can be defined as the combined volume of the dominated hypercubes domHC_r of all solution points $\mathbf{x} \in \hat{\mathcal{P}}$ regarding a reference point $\mathbf{r}$, i.e.,

$$\text{HV}_r(\hat{\mathcal{P}}) := \mu \left(\bigcup_{\mathbf{x} \in \hat{\mathcal{P}}} \text{domHC}_r(\mathbf{x}) \right)$$

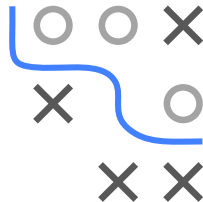
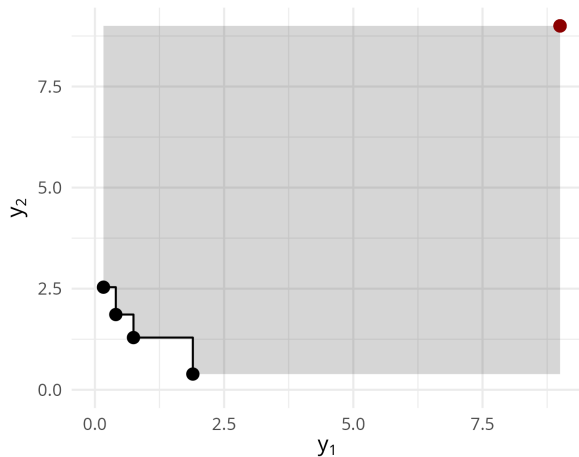
where μ is the Lebesgue measure and the dominated hypercube is given as:

$$\text{domHC}_r(\mathbf{x}) := \{ \mathbf{u} \in \mathbb{R}^m \mid f_i(\mathbf{x}) \leq u_i \leq r_i \forall i \in \{1, \dots, m\} \}$$



MULTICRITERIA BAYESIAN OPTIMIZATION

Hypervolume example:



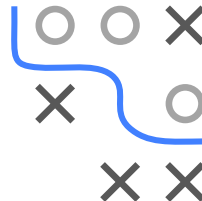
Reference point $\mathbf{r}$ in red, estimated Pareto front $\hat{\mathcal{F}}$ in black,
corresponding $HV_{\mathbf{r}}(\hat{\mathcal{P}})$ is given by the grey area

```
graph TD; subgraph Initial_design [Initial design]; A[Arbitrary DOE  
(e. g. random,  
Sobol, grid)] --> B[Individual models for  
each objective]; C[Latin Hypercube Sampling] --> D[Scalarize objectives]; end; subgraph Model_fitting [Model fitting]; B --> E[Multi-obj.  
optimization of  
infill criterion  
on each model]; D --> F[Multiple single-  
obj. optim. of  
(aggregating)  
infill criterion]; G[Model of scalarization] --> H[Single-obj.  
optimization of  
(aggregating)  
infill criterion]; end; subgraph Candidate_generation [Candidate generation]; E --> I[Select design  
point(s) with  
regard to  
other criterion]; F --> J[Select design  
point(s) with  
regard to  
infill criterion]; H --> J; end; subgraph Stopping_decision [Stopping decision]; I --> K[Decide termination  
based on approximation]; J --> L[Decide termination  
based on total budget]; K --> L; L -- Update design --> C; end;
```

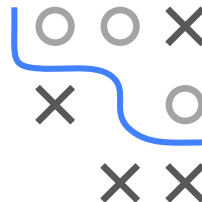
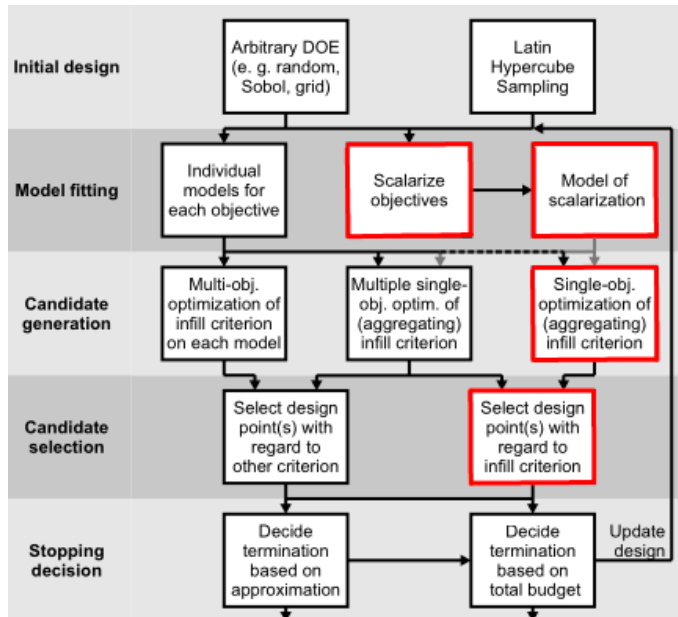
The flowchart illustrates an iterative design optimization process, organized into five main stages:

- Initial design**: Starts with two options: "Arbitrary DOE (e. g. random, Sobol, grid)" and "Latin Hypercube Sampling". Both lead into the Model fitting stage.
- Model fitting**: This stage branches into three parallel paths:
 - "Individual models for each objective" leads to "Multi-obj. optimization of infill criterion on each model".
 - "Scalarize objectives" leads to "Multiple single-obj. optim. of (aggregating) infill criterion".
 - "Model of scalarization" leads to "Single-obj. optimization of (aggregating) infill criterion".
- Candidate generation**: The outputs from the optimization steps feed into selection criteria:
 - "Multi-obj. optimization..." feeds into "Select design point(s) with regard to other criterion".
 - "Multiple single-obj. optim..." and "Single-obj. optimization..." both feed into "Select design point(s) with regard to infill criterion".
- Stopping decision**: The final selection step leads to two decision points:
 - "Select design point(s) with regard to other criterion" leads to "Decide termination based on approximation".
 - "Select design point(s) with regard to infill criterion" leads to "Decide termination based on total budget".

An "Update design" feedback loop connects the "Decide termination based on total budget" box back to the "Latin Hypercube Sampling" box in the Initial design stage.



PAREGO



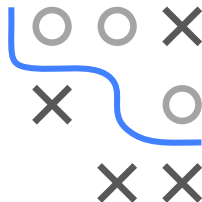
PAREGO

- 1 Scalarize standardized objectives using the augmented Tchebycheff norm

$$\max_{i \in \{1, \dots, m\}} w_i f_i(\mathbf{x}) + \rho \sum_{i=1}^m w_i f_i(\mathbf{x})$$

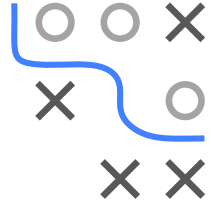
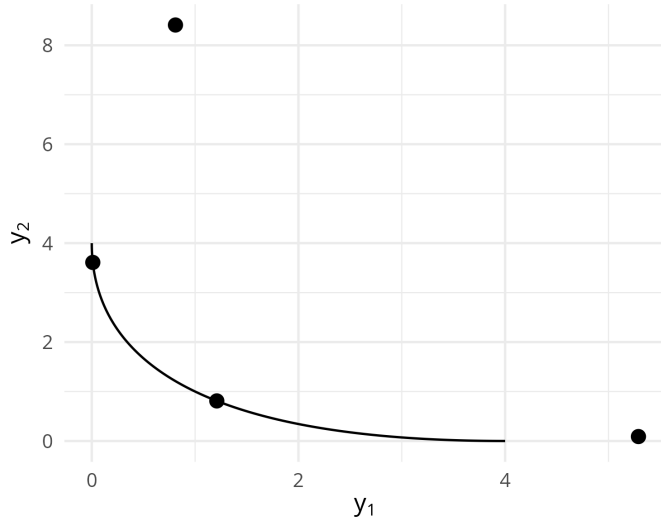
with weight vector $\mathbf{w}$ drawn uniformly from the set of evenly distributed weight vectors $\mathcal{W}$

- 2 Fit SM on the scalarized objective function
- 3 Proceed to use any standard single-objective acquisition function (EI, PI, LCB, ...)



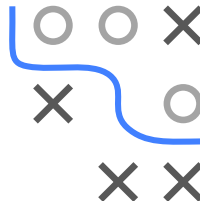
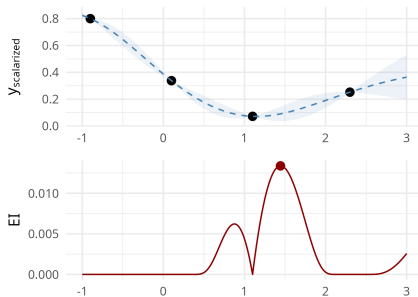
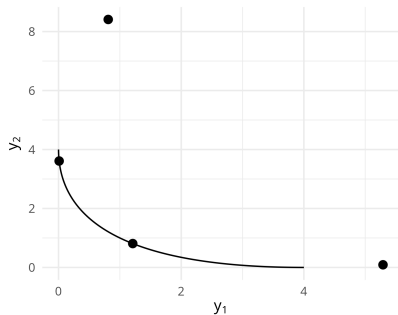
PAREGO

ParEGO Example, initial design and true Pareto front in black ...



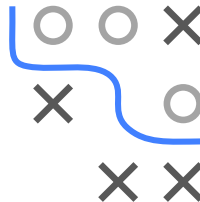
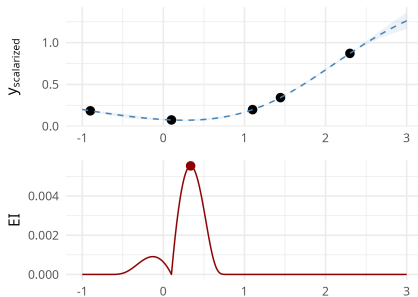
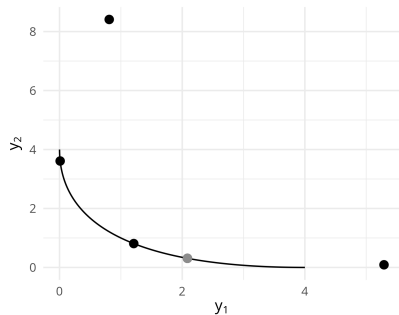
PAREGO

... standardize objectives, obtain scalarized objective via augmented Tchebycheff norm, fit SM and optimize EI ...



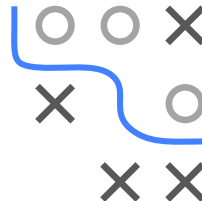
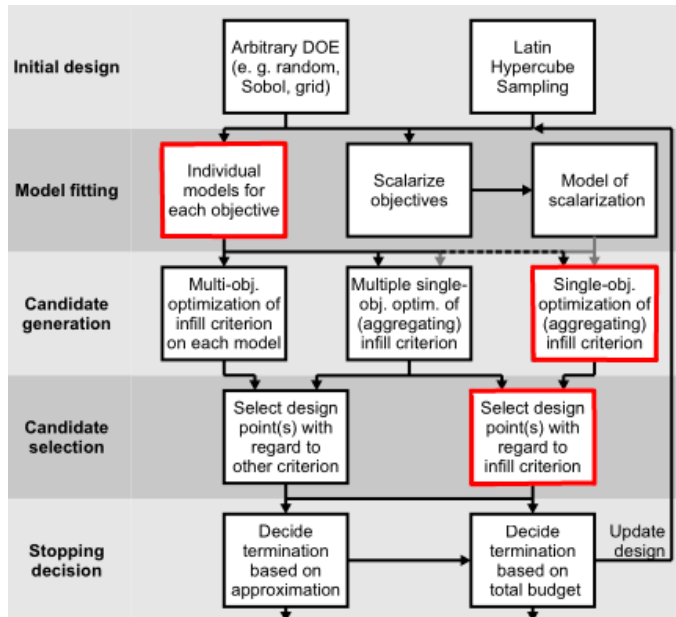
PAREGO

... note that the specific scalarization is different at each iteration!



The grey point visualizes the candidate we choose to evaluate in the previous iteration

SMS-EGO

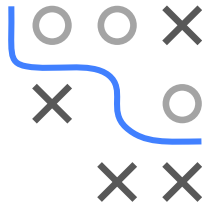
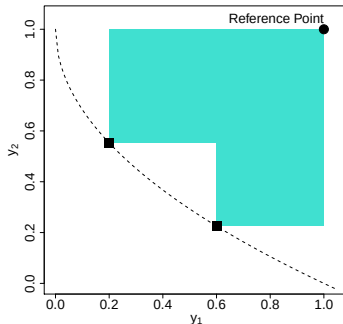


SMS-EGO

Individual models for each objective f_i

Single-objective optimization of aggregating acquisition function:
Calculate contribution of the confidence bound of candidate to the current front approximation

- Calculate LCB for each objective
- Measure contribution with regard to the hypervolume improvement
- For ε -dominated ($\prec_\varepsilon$) solutions, a penalty is added

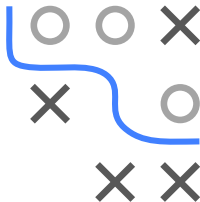
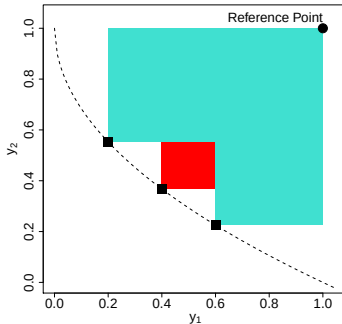


SMS-EGO

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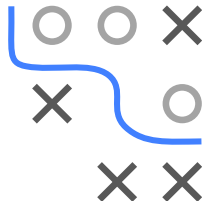
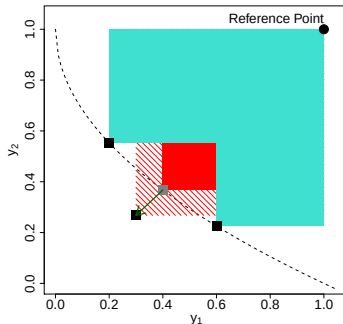


SMS-EGO

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OUTLOOK

- Many more options exist:
 - Expected Hypervolume Improvement
 - Multi-EGO
 - Entropy based: PESMO, MESMO
 - ...

